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Comprehensive review

International Liver Transplantation Society/Liver Intensive Care Group of Europe guidelines for cardiovascular assessment before liver transplantation

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Abbreviations: ACLF, acute-on-chronic liver failure; AF, atrial fibrillation; AHA, American Heart Association; AKI, acute kidney injury; ALF, acute liver failure; AST, American Society of Transplantation; AUROC, area under the curve; BNP, B-type natriuretic peptide; CABG, coronary artery bypass surgery; CACS, coronary artery calcification score; CAD, coronary artery disease; CAD-LT, coronary artery disease in liver transplantation; CAG, coronary angiography; CCM, cirrhotic cardiomyopathy; CTCA, cohort evaluated by angiograph; CPET, cardiopulmonary exercise testing; CT, computed tomography; CT-FFR, computed tomography-derived fractional flow reserve; DM, diabetes mellitus; DSE, dobutamine-stress echocardiography; ECG, electrocardiogram; ESAIC, European Society of Anesthesia and Intensive Care; ESC, European Society of Cardiology; ESLD, end-stage liver disease; FFR, fractional flow reserve; HCM, hypertrophic cardiomyopathy; hs-TnI, high-sensitivity troponin I; ICA, invasive coronary angiography; ICU, intensive care unit; IVUS, intravascular ultrasound; LOS, length of stay; LT, liver transplantation; LVOT, left ventricle outflow tract; MACE, major adverse cardiovascular events; MASH, metabolic dysfunction-associated steatohepatitis; MELD, model for end-stage liver disease; MPI, myocardial perfusion imaging; OCT, optical coherence tomography; PCI, percutaneous coronary intervention; RCT, randomized controlled trial; SAVR, surgical aortic valve replacement; TAVR, transcatheter aortic valve replacement; TR, tricuspid regurgitation; TTE, transthoracic echocardiogram.

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ABSTRACT

Cardiovascular disease is the leading cause of morbidity and mortality among liver transplantation (LT) recipients. A panel of 24 experts from the International Liver Transplantation Society and the Liver Intensive Care Group of Europe convened to develop guidelines for preoperative cardiovascular assessment of LT candidates. The panel identified 7 areas required for standard cardiovascular preoperative assessment. Quality of evidence was evaluated using the Grading of Recommendations Assessment, Development, and Evaluation approach. Strong or conditional recommendations were provided according to the overall level of evidence. Good practice statements were provided for recommendations with only low to very low quality of evidence.

After 2 rounds of deliberations, the panel provided 7 strong (high level of evidence, grade 1 ±) recommendations and 6 conditional (low level of evidence, grade 2 ±) recommendations for cardiovascular evaluation before LT. Seven recommendations were based on good practice statements because of a very low level of evidence. Agreement between panelists was reached for 100% of recommendations. Several recommendations are made to improve a candidate's cardiovascular assessment before LT.

1. Introduction

More recently, subgroups of patients with high-risk end-stage liver disease (ESLD), including acute-on-chronic liver failure (ACLF), significant frailty, and those with a high risk of coronary artery disease (CAD), are increasingly accepted for liver transplantation (LT).[1-5] Patients with ESLD with treated CAD undergoing major surgery have a similar perioperative risk compared with patients with CAD without ESLD.[1,2,6] Within the transplant community, there is an increasing understanding that cardiovascular evaluation before listing should be standardized to identify not only patients with an acceptable cardiovascular risk but also to exclude patients whose clinical condition would preclude listing.[7] Although cardiovascular disease remains the most common cause of nongraft-related mortality after LT,[8,9] cardiovascular evaluation varies substantially among centers.[10]

There are several limitations to developing standardized recommendations:

1. The pathophysiology of ESLD is characterized by a hyperdynamic circulation with low systemic vascular resistance. This can mask an underlying cardiomyopathy due to reduced contractile reserve, which is associated with low arterial elastance and impaired ventricular-arterial coupling.[11]
2. The inability to adequately increase heart rate during stress testing (*chronotropic incompetence*) limits the utility of exercise testing.
3. Decreased functional status can mask clinical cardiovascular symptoms, leading to overuse of invasive coronary angiography (ICA).[12]
4. Patients with ESLD are not well represented in general surgical risk scores. The algorithm of the Society of Thoracic Surgeons (STS-PROM score) does not take into account the severity of liver disease,[13] the European System for Cardiac Operative Risk Evaluation II (EuroSCORE II) does not include liver disease as a comorbidity,[14] and the VOCAL-Penn score has not been validated in the LT population.[15]
5. There are significant discrepancies between current guidelines for assessing patients with ESLD and general recommendations. Although the American Association for the Study of Liver Diseases/American Society of Transplantation (AST) 2013 guidelines suggest functional noninvasive tests for initial screening for all LT candidates,[16] recent American Heart Association (AHA)/AST guidelines do not support this recommendation.[17] The European Association for the Study of the Liver guidelines proposed using

anatomical imaging tests[18]; however, in 2016, their guidelines supported using stress testing.[19]
6. Most centers use locally developed CAD risk protocols, but standardized screening for valvular disease, arrhythmias, and cardiomyopathy is lacking.
7. Proper cardiovascular assessment requires a multidisciplinary team, which can pose logistical challenges.

The aim of this document is to summarize the existing literature on preoperative cardiovascular risk stratification and provide recommendations for cardiovascular assessment before LT.

2. Methods

2.1. Principle questions

These guidelines were developed by a panel of experts from the Cardiovascular Special Interest Group of the International Liver Transplantation Society and the Liver Intensive Care Group of Europe council, with designated authors assigned to specific questions.

Questions were formulated in accordance with the Patient, Intervention, Comparison, Outcome format.[20] Before literature analysis was begun, the committee agreed on the following principles:

1. A general cardiovascular evaluation *must* be performed for all LT candidates in order to reduce perioperative morbidity and mortality.
2. Specific tests incorporated into management algorithms that *should* be used according to a patient's CAD risk to reduce perioperative morbidity and mortality.
3. Minimum recommended evaluation for emergently listed LT candidates with ACLF or acute liver failure (ALF) hospitalized in the intensive care unit (ICU).
4. Specific interventions that *should* be performed on LT candidates according to their CAD risk profile in order to reduce perioperative morbidity and mortality.
5. Specific assessment and interventions that *should* be performed in LT candidates with atrial fibrillation (AF).
6. Specific preoperative assessment and interventions *should* be performed in LT candidates with valvular disease.
7. Specific preoperative assessment and interventions *should* be performed in LT candidates with cardiomyopathy.

Several areas outside the scope of the present guidelines were collectively considered by the panel:

- Guidelines for evaluating portopulmonary hypertension and right heart pathology (including pulmonary valvular disease) have been developed and published by the International Liver Transplantation Society.[21]

Posttransplant cardiovascular surveillance should be tailored to a patient's preoperative evaluation, recognizing that cardiovascular risk increases following LT.

- Combined organ transplantations were considered too specific for inclusion in this document.

2.2. Literature analysis

Literature from the past 15 years was reviewed using PubMed, EMBASE, and ClinicalTrials.gov, limited to English-language studies. Analysis followed Grading of Recommendations Assessment, Development and Evaluation (GRADE) methodology (*Grade of Recommendation Assessment, Development, and Evaluation*, https://www.gradeworkinggroup.org/),[22] with evidence levels based on study type, adjusted for quality and risk-benefit considerations.

2.3. Recommendation development

The recommendation level was binary: either positive or negative, strong, or conditional. A "high" level of evidence justified a strong recommendation (GRADE 1+/1−). A moderate or low level of evidence warranted a "conditional" recommendation (GRADE 2+/2−). When the level of evidence was very low or literature was not available, the GRADE method could not be applied, and only expert opinion was used to make "good practice statements." In the case of conflicting results between studies with moderate to high levels of evidence, the panel could not provide a recommendation.

Panelists independently rated recommendations on a 1-9 Likert scale. A recommendation was valid if ≥50% rated it ≥7 and <20% rated it ≤3. Strong agreement was made if at least 70% of the participants had a convergent opinion (≥7). If no agreement was reached, recommendations were revised and rerated until a consensus was reached.

Question 1. What general (non-LT-specific) cardiovascular assessment must be performed in all LT candidates in order to reduce perioperative morbidity and mortality?

3. Results

3.1. Minimal cardiovascular examination

Electrocardiography (ECG) and transthoracic echocardiography (TTE) are recommended by the *European Society of Cardiology (ESC)/European Society of Anesthesiology and Intensive Care* before major noncardiac surgery in patients with suspected cardiovascular disease or unexplained signs or symptoms.[23] Observational studies suggest both ECG and TEE influence perioperative outcomes.[24-26]

AASLD recommends ECG and TTE for all LT candidates.[16,27] There are no randomized controlled trials (RCTs) validating the role of ECG or TEE for LT. ECG is used as a screening tool to detect unknown arrhythmia or ischemia/myocardial infarction. Corrected QT interval prolongation occurs in 33% to 65% of ESLD patients[28] and predicts mortality after model for end-stage liver disease (MELD) adjustment; however, this was not confirmed in LT patients.[29-31]

In LT candidates, resting TTE should be performed to rule out conditions affecting post-LT outcomes (see question 7).[32,33]

Recommendation 1.1.

The panel recommends performing ECG and TTE in all LT candidates on the waiting list (strong recommendation GRADE 1+, strong agreement).

3.2. Preoperative iron and hemoglobin assessment and management

Preoperative anemia affects up to 60% of patients with ESLD. Preoperative anemia (defined as a Hb level<13 g/dL) is associated with higher transfusion needs, greater risk of acute kidney injury (AKI), prolonged postoperative ventilation, longer ICU stay, and reduced graft and patient survival.[34-40] Whether low preoperative hemoglobin reflects the severity of liver disease or independently predicts poor outcomes after LT remains debated.[34,35] Several RCTs and meta-analyses demonstrate that treating iron deficiency before major surgery increases hemoglobin levels and lowers transfusion requirements without increasing rates of adverse events. Its benefit in LT recipients remains unclear.[41]

Recommendation 1.2.

The panel suggests that optimizing preoperative iron levels and hemoglobin concentration in patients on the waiting list can potentially reduce the need for allogenic blood transfusion during LT. (good practice statement, strong agreement)

3.3. Preoperative outcome markers

Biomarkers, including B-type natriuretic peptide (BNP) and high-sensitivity troponin I (hs-TnI), have been shown to be predictive of postoperative outcome after high-risk surgical procedures. ESC/*European Society of Anesthesiology and Intensive Care* recommend BNP, NT-proBNP, and hs-TnI testing before intermediate- or high-risk noncardiac surgery in patients with cardiovascular risk factors, known CAD, or CAD symptoms.[23] Cirrhotic patients with a MELD <25 and BNP <155 had 100% survival after LT, whereas those with MELD >25 and BNP >155 experienced a 27% mortality rate.[42] However, postoperative deaths were due to sepsis-related multiorgan failure with no cardiac-specific causes reported. This suggests that BNP reflects overall disease severity rather than cardiac-specific risk. The ability of BNP to predict short-term outcomes and stratify mortality risk in LT recipients was confirmed by Chung et al.[43] The BNP value demonstrated an outstanding predictive ability for 1-month mortality after LT (area under the curve [AUROC] = 0.976), compared with the MELD and MELD-Na scores (AUROC = 0.714 and 0.690, respectively). In an Asian LT registry including 3811 patients, increasing BNP during the first 3 postoperative days was also strongly associated with increased mortality.[44]

The impact of troponin on LT outcomes remains unclear. Initial findings demonstrating an association of troponin I levels >0.07 and cardiac events[45] have not been confirmed by recent studies.[46,47]

Elevated post-LT troponin I and BNP levels may reflect nonspecific factors such as AKI or sepsis and therefore should be interpreted with caution. Although the AHA/ESC guidelines recommend assessment of BNP and troponin during preoperative evaluation, specific evidence in LT remains limited.

Clinical decisions should rely on both biomarker values and the severity of ESLD, along with other contributing factors. Moon et al[48] investigated the combined prognostic value of hs-TnI and BNP levels in a large retrospective cohort of 2490 patients. Ninety-day mortality was 1% when both BNP <400 pg/mL and hs-TnI <60, whereas 9% if either was elevated and 19.4% if both were increased above these thresholds.[48] Further stratification showed that patients with MELD >15 and elevated biomarkers had a 90-day mortality of 19.9%, compared with 1.7% when both biomarkers were low.

These results reinforce the recommendation that assessment should integrate multiple cardiac biomarkers, with particular emphasis on patients presenting with higher MELD scores, where the risk of cardiac complications is greater.

Recommendation 1.3.

The panel suggests that preoperative BNP and hs-TnI measurements may help risk stratify early post-LT mortality. (good practice statement, strong agreement)

3.4. Functional status assessment

Physical capacity and frailty should always be assessed in LT candidates. Frailty, represented by reduced endurance, balance, activity, and ability to walk, can be evaluated using cardiopulmonary exercise testing (CPET), handgrip strength, Karnofsky Performance Status, and 6-minute walk distance.

Several studies have demonstrated that 6MWD is an independent predictor of waitlist mortality.[49-51] In the most recent study, with a threshold of 390 ± 101 m, there was a 29% increase in risk of death for each 50-m decrement in this value, which was independent of age, gender, BMI, or MELD-Na score.[49]

Handgrip strength and CPET were associated with both increased post-LT hospital and ICU length of stay (LOS).[52,53] A recent review supports CPET as an objective and independent predictor of both pre- and posttransplant mortality. Among included studies, median baseline peak VO_2 values and anaerobic threshold ranged from 14.2 to 20.8 mL/kg/min and from 11.3 to 12.4 mL/kg/min, respectively. However, substantial heterogeneity in study designs, particularly in the timing of testing, CPET variables assessed, and cutoff values, precluded the establishment of definitive CPET mortality thresholds.[54]

Reduced physical capacity, regardless of assessment method, is associated with higher post-LT disability and mortality.[55-57] In 307 ambulatory ESLD patients, Duke Activity Status Index and Liver Frailty Index assessments demonstrated that 81% of patients were prefrail or frail.[58] Lai et al[59] found that frail LT recipients had higher post-LT mortality and health care utilization than nonfrail patients. Pre-LT frailty was also found to be associated with worse global functional health 1 year after LT.[60] Analysis of 2851 visits from 1093 outpatients with cirrhosis demonstrated that frailty improvement reduced the risk of death or delisting, supporting ongoing frailty monitoring and prehabilitation efforts.[61] It is very difficult to determine a specific level of frailty precluding LT. Rather, frailty should be integrated as one of several key components in a comprehensive global

assessment. Moreover, a recent study demonstrated a survival benefit of LT regardless of degree of frailty.[62]

Because low muscle mass is a key element of frailty, the psoas muscle index is considered a surrogate or objective marker of physical frailty, particularly in patients undergoing major surgery or those with end-stage organ disease or heart failure. It has been demonstrated that a low psoas muscle index is associated with both a higher incidence of cardiovascular complications and increased overall postoperative morbidity.[63]

Recommendation 1.4.

The panel recommends frailty and physical capacity assessment before LT to predict perioperative morbidity and mortality and identify patients who would benefit from prehabilitation (strong recommendation GRADE 1+, strong agreement).

Question 2. Are there specific cardiovascular tests that should be used according to a LT candidate's CAD risk to reduce perioperative morbidity and mortality?

The utility of cardiovascular screening modalities to predict post-LT cardiovascular events and death has been assessed in multiple studies (Table 1).[64-73] The vast majority of available studies had limitations:

1. Most are retrospective, single-center evaluations with a relatively low number of patients, with the potential for significant selection bias.
2. Variable CAD definitions or criteria are used.
3. Therapeutic interventions performed either before or after LT are not always included in outcome assessment.

3.5. Clinical scoring

Several studies assessed the ability of nonspecific cardiovascular risk scores to detect asymptomatic CAD and predict post-LT cardiovascular events.[74-76] All scores indicated that more risk factors correlate with significant CAD and worse post-LT outcomes. Two LT-specific scores are currently available: the cardiovascular risk in orthotopic LT[64] and the coronary artery disease in liver transplantation (CAD-LT) (Table 2).[12] The former was developed to predict the incidence of 1-year post-LT cardiovascular events and has been assessed in 3 external cohorts of 467, 158, and 273 patients. Cardiovascular risk in orthotopic LT demonstrated a relatively poor predictive ability with AUROCs ranging between 0.62 and 0.72.[65,66] The CAD-LT (Table 2) demonstrates an AUROC of 0.76.[12] In the LT population, it would have been able to detect 97% of patients with significant CAD. If the CAD-LT score had been used, 44% of ICA and 57% of stress echocardiograms could have been avoided.[12] A study evaluating a modification of the CAD-LT (without a family history of CAD) in 120 LT candidates demonstrated an AUROC of 0.722 for CAD.[67] Similarly, CAD-LT has been found to predict significant CAD in a cohort evaluated by CTCA.[65,67,68] These results support the use of the CAD-LT score to estimate risk and tailor preoperative examinations as proposed in Table 2. None of these scores includes patients with metabolic dysfunction-associated steatohepatitis (MASH). MASH has been shown to be associated with major adverse cardiovascular events (MACE) and obstructive CAD independent of the presence of diabetes or metabolic disease.[77]

3.6. Stress testing

Stress tests have been recommended as a screening tool by several international guidelines,[16,19] although they are no longer recommended in the most recent EASL statement.[18] Several studies evaluated the diagnostic and predictive value of dobutamine-stress echocardiography (DSE), myocardial perfusion imaging (MPI), single-photon emission computed tomography, and positron emission tomography for diagnosing CAD in LT candidates.[69,70] For diagnosing CAD, sensitivity was 25% to 29% for DSE and 62% for MPI, whereas specificity was 68% to 87% and 60%, respectively.[69,70,78,79] The sensitivity of DSE remained low even when patients achieved their target heart.[78,79] Studies could not consistently demonstrate an association between stress tests and post-LT cardiovascular events, and the estimation of risk was limited by very large confidence intervals.[80-82] A positive DSE or MPI has limited accuracy in predicting post-LT mortality.[69] Cardiac magnetic resonance imaging may predict cardiac outcomes after LT.[83,84]

3.7. Anatomical imaging

Most available studies used computed tomography (CT) imaging techniques, including CT coronary angiography CTCA and coronary artery calcification score (CACS).[67,71,72,85-91] A recent review, including 17 studies, concluded that CT-detected vascular calcifications are significantly associated with both the presence and severity of CAD. Calcification in the descending aorta and aortic root demonstrated a stronger correlation with CAD than calcification in the aortic arch or ascending aorta. However, none of the studies evaluated patients with liver disease.[92]

CT-based tests show high sensitivity and negative predictive value but low specificity, suggesting potential CAD overestimation. Because of the ability of both CACS and CTCA to predict post-LT short and long-term cardiovascular events,[67,68,72,73,86,87,90,91] these techniques could be combined.[11] Given its strong negative predictive value, further assessment is not necessary if the CACS is zero.[71,89,93] ICA is indicated if the calcium score exceeds 400 (predicting the need for revascularization and increased risk of early complications after LT). Contrast-enhanced CTCA assesses atherosclerosis, quantifies stenosis, and distinguishes plaque types. Noncalcified or mildly calcified plaques are rupture-prone, whereas heavily calcified ones are typically stable.[89,93,94] A calcium score between 0 and 400 (units) should prompt a CTCA (Fig. 1). The utility of CT-based assessments is limited in patients with a history of coronary stents or coronary artery bypass surgery (CABG).[95] Anatomical testing may not be available in some resource-constrained settings. In such situations, careful cardiovascular risk assessment becomes even more critical, particularly with the use of clinical risk scores such as CAD-LT. When available, stress testing may be performed; however, a negative result does not reliably exclude significant CAD.

Table 1
Ability of available screening modalities (clinical scores, stress tests, and anatomical imaging) to diagnose coronary artery disease and predict postoperative cardiovascular events.

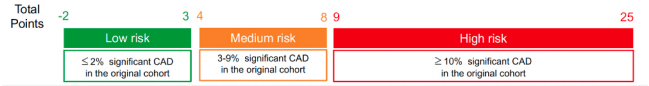
Screening modalities	Name of the test	Coronary artery disease diagnosis					Post-LT-event prediction						References
		Se	Sp	PPV	NPV	AUROC	Se	Sp	PPV	NPV	AUROC/c-statistic	Other predictive characters	
Clinical scores	CAR-OLT	N/A	N/A	N/A	N/A	N/A					0.62-0.78		64-66
	CAD-LT (mCAD-LT)[a]	95-86[a]	28-50[a]	42[a]	89-94[a]	0.722[a]-0.76	48[a]	79[a]	30[a]	79[a]	0.64		65,67,68
Functional tests	Dobutamine-stress echocardiography	25	68								0.639	RR 30.2 (2.8-325.4)	69,70
	Myocardial perfusion imaging	62	60									RR 2.6 (1.09-6.1)	69,70
Anatomical tests	Coronary artery calcification score-CACS	52-80	53-87	30	74	0.502-0.82	52	84	44	88			67,71,72
	CT coronary angiography-CTCA	100	31	34	100	0.654	62	75	23	94-100			67,73

AUROC, area under the curve; CACS, coronary artery calcification score; CAD-LT, coronary artery disease in liver transplantation; CAR-OLT, cardiovascular risk in orthotopic liver transplantation; CT, computed tomography; LT, liver transplantation; N/A, not applicable; NPV, negative predictive value; PPV, positive predictive value; Se, sensitivity; Sp, specificity.

[a] mCAD-LT, modified CAD-LT.

Table 2
CAD-LT score (adapted from Rachwan et al).[12]

Points associated with each factor	
Factor	**Points**
Age	
<30	0
30-39	2
40-49	4
50-59	6
60-70	8
>70	10
Sex	
Male	0
Female	−2
Diabetes	
Yes	2
No	0
Hypertension	
Yes	2
No	0
Tobacco (pack-years)	
0-20	0
21-40	1
>40	2
Family history of CAD	
Yes	2
No	0
Personal history of CAD	
Yes	7
No	0



CAD, coronary artery disease; CAD-LT, coronary artery disease in liver transplantation.

3.8. Invasive CAG

ICA is considered the reference standard for diagnosing CAD.[6,12,69,96-99] Indications for ICA vary significantly among LT centers. Therapeutic procedures performed (or not) from ICA findings are a significant confounding factor, which is difficult to account for. Even in highly selected patient cohorts having ICA after noninvasive tests, the prevalence of significant CAD is low (<20% to 30%). Some studies suggest a trend for MACE according to the extent of CAD found with ICA, but this trend is not statistically significant in most studies.[6,69,97,100]

3.9. Fractional flow reserve

Fractional flow reserve (FFR) measures the ratio of blood flow in a diseased coronary artery to the maximum flow in a normal artery. In stable CAD, FFR is the best tool to guide percutaneous coronary intervention (PCI) for significant stenoses.[101,102] Initially, routinely measured during ICA, FFR is now performed noninvasively during CTCA and MPI.[95,103] FFR increases the specificity of CTCA in the evaluation of CAD and helps plan revascularization.[95] Patients with intermediate-risk coronary anatomy with CTCA (ie, presence of luminal stenosis between 50% and 70%[17] and an abnormal FFR can undergo ICA and revascularization, whereas for those with normal FFR, ICA can be deferred. In a general population of 208 patients with suspected CAD, the diagnostic accuracy of FFR was higher with positron emission tomography compared with single-photon emission computed tomography and CTCA.[103] One retrospective study evaluated the accuracy of a computed tomography-derived FFR (CT-FFR) machine learning algorithm in 67 LT candidates with several coronary lesions deemed significant by CTCA and who underwent ICA.[104] Compared with ICA, sensitivity, specificity, positive predictive value, and negative predictive value of

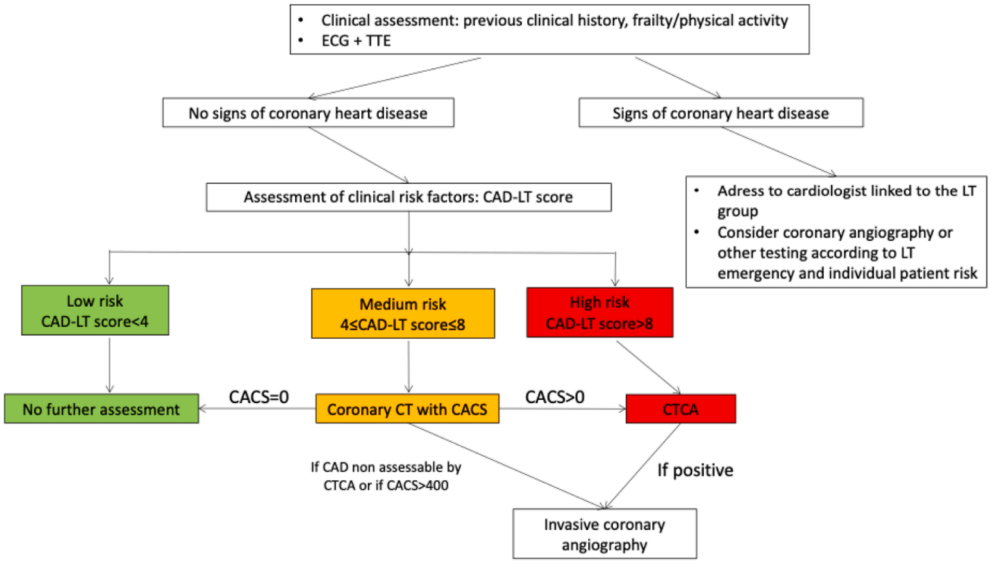


Figure 1. Algorithm proposed by the experts for coronary artery disease risk assessment in asymptomatic patients. CACS, coronary artery calcification score; CAD, coronary artery disease; CAD-LT, coronary artery disease in liver transplantation; ECG, electrocardiogram; LT, liver transplantation; TTE, transthoracic echocardiogram.

CT-FFR were 71% (49% to 92%), 90% (82% to 98%), 67% (45% to 88%), and 91% (84% to 99%), respectively. Machine learning-based CT-FFR appears to be a promising noninvasive approach to evaluate patients for hemodynamically significant coronary stenoses in patients evaluated for LT and is likely to reduce the use of ICA in high-risk populations.

3.10. Other novel techniques

In recent years, both intravascular ultrasound (IVUS) and optical coherence tomography (OCT) have been increasingly used to optimize the use of PCI, particularly with complex CAD. IVUS provides greater tissue penetration and is preferred for assessing vessel size, plaque burden, and stent expansion, particularly with large or heavily calcified vessels. OCT, on the other hand, offers superior resolution for visualizing plaque microstructure and detailed stent–vessel interactions. Both modalities have been shown to improve clinical outcomes compared with angiography alone and are recommended in selected cases. The choice between IVUS and OCT depends on patient-specific factors such as vessel size, renal function, and the need for contrast.[54]

To reduce perioperative morbidity and mortality due to CAD-related events, the panel recommends the following:

Recommendation 2.1

The panel endorses the AHA-AST Scientific Statement and recommends clinically evaluating CAD risk in LT candidates by assessing the cumulative presence of cardiovascular risk factors (including MASH) to (strong recommendation GRADE 1+, strong agreement).

Recommendation 2.2

The panel recommends using the CAD-LT score as a first-line assessment to screen candidates and identify recipients at the highest risk of post-LT events. Further tests should be performed on LT candidates with a high risk for CAD (conditional recommendation GRADE 2+, strong agreement).

Recommendation 2.3

The panel recommends using CT-based tests (ie, CTCA and CACS) to identify LT candidates with a higher risk of significant CAD identified using the CAD-LT score (conditional recommendation GRADE 2+, strong agreement).

Recommendation 2.4

Given their limited value in diagnosing CAD in LT candidates, the panel does not recommend performing stress tests as a first-line assessment (strong recommendation GRADE 1+, strong agreement).

Recommendation 2.5

The panel recommends performing ICA according to estimated risk based on noninvasive tools (limiting lesions on CTCA, CACS >400) or in patients with previous PCI or CABG. Routine use of ICA is not recommended (strong recommendation GRADE 1+, strong agreement).

Recommendation 2.6

The panel suggests using the algorithms proposed in Figure 1 to assess CAD risk (good practice statement, strong agreement).

Question 3. What is the minimum recommended cardiac evaluation for LT candidates hospitalized in the ICU with ACLF or other acute forms of liver failure who are emergently listed?

Due to high ACLF waitlist mortality, early LT offers the best chance for survival.[105,106] In this complex setting, a moderately increased risk of perioperative cardiac complications should be considered. ACLF and ALF often cause significant hemodynamic instability from systemic inflammation and sepsis.[107] Resuscitation attempts can lead to fluid overload, pulmonary edema, and significant cardiac dysfunction.[108] Adequate hemodynamic monitoring and continual re-evaluation are essential for all LT candidates with ACLF/ALF in the ICU.[109] Noninvasive monitoring is increasingly used to diagnose and manage hemodynamic disturbances in patients with ESLD.[110] Although complications of invasive hemodynamic monitoring are low,[111] it should be reserved for selected patients.[109] Because the majority of patients with ACLF were not previously listed as LT candidates, TTE is considered a basic, mandatory initial screening tool. If cardiac risk is low to moderate based on risk factors and echocardiography, proceeding with LT is reasonable, whereas delays could miss a narrow window of opportunity.

If patients have a high probability of significant CAD (CAD-LT score >8), an ICA should be considered, weighing the pros/cons of delaying a life-saving LT.[112] If a treatable lesion is found, decisions on stenting, angioplasty, or antiplatelet therapy should weigh cardiac benefits against the risk of delaying LT on a case-by-case basis.

If a thoraco-abdominal CT scan is performed during the ICU stay as part of a pre-LT evaluation (eg, for preoperative screening of abdominal vessels), coronary arteriography can be performed at the same time.

Recommendation 3.1

For patients with ALF and ACLF hospitalized in the ICU and emergently listed for LT, a cardiovascular risk assessment should be performed to include an electrocardiogram, TTE, and cardiac biomarkers (good practice statement).

Recommendation 3.2.

The panel further suggests screening following the algorithm proposed in Figure 2, considering the risks of proceeding with LT in the face of significant CAD vs delaying a life-saving LT (good practice statement, strong agreement).

Question 4. Are there specific interventions that should be performed before LT in patients with CAD in order to reduce perioperative morbidity and mortality?

Of all potential pretransplant interventions, only PCI and CABG for obstructive CAD (>50% stenosis of a major coronary vessel) have been evaluated. Posttransplant all-cause mortality and MACE were determined as outcomes in these evaluations.

In asymptomatic patients, although recent data suggest that ICA is relatively safe in LT candidates,[113] there is a lack of robust evidence supporting the routine use of coronary revascularization for obstructive CAD.[114,115] However, coronary revascularization is suggested by current guidelines for LT candidates with high degrees of stenosis (>70%).[17,116]

No RCTs have specifically addressed this issue in LT patients. A retrospective multicenter study evaluated outcomes in

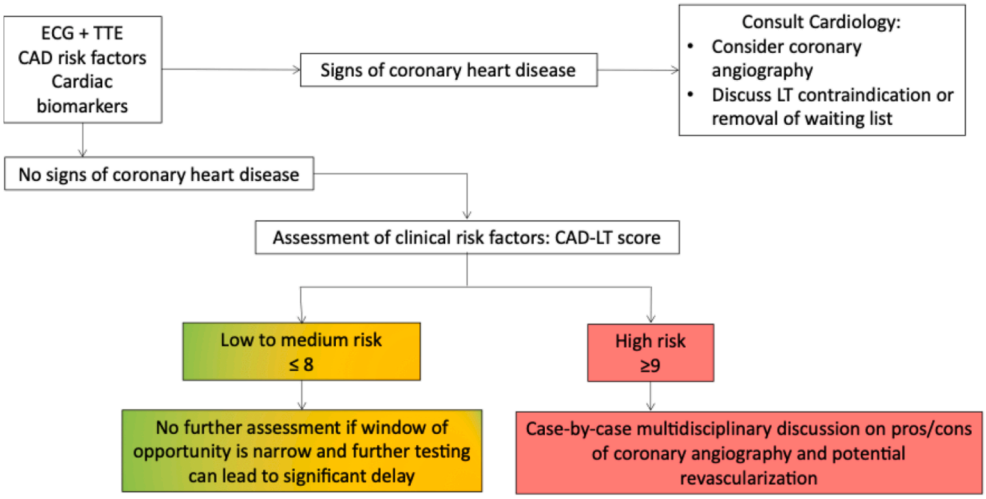


Figure 2. Algorithm proposed by the experts for rapid coronary artery disease risk assessment in asymptomatic critically ill patients who are emergently listed. CAD-LT, coronary artery disease in liver transplantation; ECG, electrocardiogram; LT, liver transplantation; TTE, transthoracic echocardiogram.

630 LT recipients who underwent pre-LT ICA; 151 had obstructive CAD (>50% stenosis).[6] Post-LT survival did not differ significantly between patients with or without obstructive CAD, regardless of revascularization (Hazard Ratio (HR), 1.42; CI, 0.95-2.12; and 0.67, CI, 0.39-1.17). These findings suggest that with appropriate pre-LT CAD management, posttransplant survival remains comparable.

Maddur et al[98] retrospectively evaluated 1221 LT patients over 3 periods (A, 2000-2004; B, 2005-2008; and C, 2009-2010). The rate of ICA increased significantly during the study periods: 15% (78/527) in period A, 52% (272/520) in period B, and 68% (119/174) in period C. PCI rates also increased over the study periods from 1%[7] in period A, 7%[36] in period B, to 8%[14] in period C. One-year mortality rate decreased over the study periods from 16% (83/527) in period A, 10% (52/520) in period B, to 6% (11/174) in period C ($P < .001$). The incidence of myocardial infarction at 1 year decreased from 8.0% in period A to 3.6% in period B and decreased to 0.6% in period C. The authors suggest that improved outcomes over time may be attributed to the increasing use of catheterization and PCI.

Satapathy et al[117] identified 87 LT patients who underwent pretransplant ICA, of whom 29 had obstructive CAD (≥50% stenosis). Twenty-two of those underwent revascularization. MACE at 90 days did not differ between those with and without obstructive CAD (8/29, 27%), (15/58, 26%, $P = .7$) respectively, nor did overall survival. In another retrospective study, Snipelisky et al[118] reported that pretransplant revascularization (PCI or CABG) did not improve outcomes.

Recommendation 4.1

The panel was not able to make a recommendation on the impact of CAD revascularization prior to LT in asymptomatic patients with obstructive CAD on mortality or MACE (no recommendation, strong agreement).

Recommendation 4.2

The panel suggests having a multidisciplinary discussion to decide whether CAD revascularization should be performed and the best timing (good practice statement, strong agreement).

Question 5. Are there specific perioperative assessments or interventions that should be performed in LT candidates with either a history of or currently have AF?

All available studies evaluating the incidence of AF were retrospective and included cohorts between 700 patients in a single institution and 45,000 in national databases. In patients with ESLD, the prevalence of AF ranges between 5% and 12.2%[119-121] and about 5.4% to 6.5% of LT candidates present with AF.[121-123] High MELD score, sleep apnea, valvular disease, hemodynamic instability, and low ejection fraction were predictors for the development of perioperative AF in LT recipients.[119,124-126]

Pre-existing or new-onset AF at the time of LT was associated with a higher risk of cardiovascular events, stroke, AKI, graft failure, longer hospital and ICU stays, and increased mortality.[120,122-133] There are not any prospective studies or guidelines evaluating or managing AF in LT candidates to reduce risk. General statements included "thorough cardiac assessment."[119,123] "careful risk stratification,"[122,130,134] "consideration for anticoagulation when there are no contraindications,"[135-137] and "development of a risk index for predicting postoperative AF."[126] In patients with cirrhosis with AF, anticoagulation was associated with reduced stroke risk without significantly increasing bleeding risk.[136,138,139] All studies highlighted the importance of individualized patient assessment.[140]

Recommendation 5.1

The panel recommends managing perioperative AF in the same fashion as in any other surgical setting: standard medical treatment, including anticoagulation and appropriate treatment

of the underlying causes (strong recommendation, GRADE 1+, strong agreement).

Recommendation 5.2

The panel suggests careful risk stratification and developing a preoperative intervention strategy on a case-by-case basis in LT candidates with pre-existing AF (best practice statement, strong agreement).

Question 6. Are there specific perioperative assessments or interventions that should be performed in LT candidates with either a history of or currently have valvular disease to reduce perioperative morbidity and mortality?

No RCTs exist for valvular heart disease treatment in patients with ESLD, but it is reasonable to apply the same criteria as with patients in the general population.[141,142] Perioperative mortality and complication rates in patients with cirrhosis increase with the severity of liver disease. Reported mortality rates are unacceptably high even in patients with Child-Pugh class A and B cirrhosis (0% to 11% and 18% to 50%, respectively), and reach 67% to 100% in patients with Child-Pugh class C cirrhosis.[143] Although outcomes of cardiac surgery in Child-Pugh class A patients may not differ significantly from those in patients without liver disease, patients in class C are likely inoperable. In patients with Child-Pugh class B cirrhosis—and even more so in class C—surgery should only be considered with the utmost caution, following thorough assessment of surgical risk and evaluation of alternative treatment modalities. However, neither Child-Pugh nor MELD scores fully account for the unique complexity of valvular heart surgery. These indices, therefore, should be integrated into a cardiac surgery-specific risk model such as the EUROSCORE as proposed by Ostovar et al.[144] Notably, combined LT and cardiac surgery procedures have been reported in only a few small case series. In high-volume centers, these complex surgeries may be feasible, with acceptable outcomes observed in highly selected patients who were otherwise considered inoperable. However, these findings require further confirmation in larger evaluations.[145]

3.11. Aortic valvular disease

In a recent meta-analysis of observational studies performed by Ndunda et al,[146] transcatheter aortic valve replacement (TAVR) was compared with surgical aortic valve replacement (SAVR) in patients with ESLD. In these patients, TAVR was associated with lower odds of in-hospital mortality (OR, 0.51; CI, 0.27-0.98), blood transfusion (OR, 0.36; CI, 0.21-0.60), and a shorter LOS (10.9 vs 15.7 days for SAVR). Similar findings (a lower rate of blood transfusion and shorter hospital LOS) were found in the study derived from the US National Inpatient Sample of patients with ESLD who underwent TAVR vs SAVR.[147,148] The authors failed to demonstrate differences in in-hospital mortality. TAVR was associated with a higher rate of postprocedure heart block requiring permanent pacemaker placement.

3.12. Mitral valvular disease

Although unstudied in cirrhosis, recent guidelines recommend percutaneous mitral commissurotomy for symptomatic or pulmonary hypertension-related mitral stenosis to temporarily reduce pressure gradients.[142]

For mitral regurgitation, only 1 recent retrospective study focused on patients with cirrhosis, comparing 123 transcatheter edge-to-edge repair and 750 surgical mitral valve repairs or replacements.[149] Patients managed with transcatheter edge-to-edge repair had lower in-hospital mortality (8.2% vs 16%, $P = .04$), transfusion requirement (30.3% vs 61.2%, $P = .02$), and fewer days on mechanical ventilation (1.2% vs 17.2%, $P = .02$).

3.13. Tricuspid valvular disease

No studies have compared treatments for tricuspid valve problems in patients with cirrhosis. In patients without cirrhosis, tricuspid valve repair improves liver function at 6 months and lowers surgical risk in patients with significant tricuspid regurgitation (TR).[150] In a propensity score matched study with no available data regarding patient liver function, transcatheter valve intervention was compared with usual medical treatment and found that transcatheter valve intervention was associated with both lower 1-year mortality (23% ± 3% vs 36% ± 3%; $P = .001$) and rate of rehospitalization (26% ± 3% vs 47% ± 3%; $P < .0001$).[151]

TR in patients with ESLD is frequently associated with portopulmonary hypertension or volume overload, although severe TR can reflect primary valvular disease. About 6% of patients with cirrhosis and portopulmonary hypertension demonstrate significant TR. Eligibility for transplant of this group of patients should be carefully evaluated.[21,152]

Recommendation 6.1

The panel suggests that indications for valvular interventions in patients with cirrhosis should be the same as for those followed in the general population (best practice statement, strong agreement).

Recommendation 6.2

The panel recommends TAVR, when feasible, over surgical aortic replacement to reduce mortality and complications in patients with severe aortic stenosis on the waiting list (strong recommendation, GRADE 1+, strong agreement).

Recommendation 6.3

The panel suggests percutaneous mitral valve repair over mitral valve surgery to reduce mortality and complications in patients with cirrhosis in combination with severe primary mitral regurgitation or severe symptomatic secondary mitral regurgitation after medical treatment (conditional recommendation, GRADE 2+, strong agreement).

Question 7. Are there specific perioperative assessments and interventions that should be performed in LT candidates with cardiomyopathy to reduce morbidity and mortality?

3.14. Hypertrophic cardiomyopathy

Hypertrophic cardiomyopathy (HCM) is defined by a left ventricular wall thickness of ≥15 mm in 1 or more cardiac segments.[153,154] HCM may cause left ventricular outflow tract (LVOT) obstruction, increasing perioperative cardiac risk.[155] HCM data in LT are limited to small retrospective studies.[32,156]

LT candidates with a diagnosis of HCM should receive preoperative TTE and formal cardiology referral.[17] Although it is a strong risk factor for CAD,[17] HCM by itself should not be a contraindication to LT. However, if associated with a higher resting LVOT gradient, HCM may increase the risk of post-LT mortality and morbidity. A resting gradient greater than 60 mmHg may be considered a relative contraindication to LT.[32,156] Inducible LVOT obstruction (provoked by isoproterenol challenge, dobutamine-stress test, or detected after Valsalva maneuver) without cardiopulmonary symptoms is likely not a contraindication to LT.[157] Intraoperatively, it is beneficial to use TEE to assist with fluid management and real-time detection of LVOT obstruction and systolic anterior motion of the mitral valve. The role of medical and septal reduction therapies in HCM before LT is unclear and should be evaluated on a case-by-case basis.

3.15. Cirrhotic cardiomyopathy

Cirrhotic cardiomyopathy (CCM) results from a specific change in the myocardium and affects both systolic and diastolic function. Some retrospective studies suggest CCM is associated with poorer outcomes and increased cardiovascular events during and after LT. It has been demonstrated that CCM impacts post-LT outcomes independent of traditional cardiac risk factors.[33,158,159]

A new definition of CCM (2020) is based on echocardiographic findings with criteria according to the CCM Consortium (Table 3).[108] It has been shown that patients who meet the new diagnostic criteria for CCM have nearly double the risk of experiencing MACE after LT, with a reported HR of approximately 1.93.[158]

Table 3
Diagnostic criteria for cirrhotic cardiomyopathy (adapted from Izzy et al).[108]

Systolic dysfunction	Diastolic dysfunction (at least 3[a])
Left ventricular ejection fraction (<50%)	Left atrial volume index >34 mL/m^2
Global longitudinal strain absolute value of <18%	Septal e' <7 cm/second or lateral e' <10 cm/second
	Tricuspid regurgitation peak velocity >2.8 m/second
	E/e' ratio ≥15

[a] Patients with 2 criteria only need further evaluation using left atrial strain, E/A on Valsalva, and assessment of pulmonary veins.

Critical differences have been highlighted when comparing these revised TTE-based diagnostic criteria with the original 2005 Montreal criteria.[160,161] In the study by Spann et al,[158] 30% of 210 patients met the revised criteria prior to LT, as compared with 77% with the original criteria. Among the revised CCM criteria, low septal e' was the most predictive variable for MACE after LT (HR, 3.45; $P < .001$).[158] Left atrium enlargement and increased E/e' greater than 10 were also independent predictors of poor outcome in several cohorts of patients with cirrhosis,[162,163] including cardiac decompensation after transjugular intrahepatic portosystemic shunt.[163] Other TEE findings including systolic dysfunction and hypertrophic cardiac remodeling[164] were described as associated with reduced survival after LT. A recent meta-analysis evaluated 13 studies, including a total of 802 patients with cirrhosis and 540 healthy controls. The authors found that global longitudinal strain can serve as an early marker of myocardial dysfunction in patients with cirrhosis.[165] In addition, it has been demonstrated that less negative resting global longitudinal strain values may help identify liver transplant candidates with latent systolic dysfunction.[166]

CCM associated with significantly reduced LV function is a contraindication for LT, although no clear cutoff for ejection fraction or diastolic dysfunction exists. If LT proceeds despite significant LV dysfunction, perioperative mechanical circulatory support must be available as rescue therapy. Patients with CCM may benefit from perioperative echocardiographic monitoring.

In patients with CCM with diastolic dysfunction, the use of sodium–glucose cotransporter-2 inhibitors, such as empagliflozin or dapagliflozin, may present a promising therapeutic option. However, this approach should be considered only in carefully selected patients with hemodynamic stability and adequate renal function. Importantly, the efficacy and safety of sodium–glucose cotransporter-2 inhibitors in populations of patients with cirrhosis require confirmation through clinical studies.[167] Angiotensin receptor–neprilysin inhibitors may be considered part of antiremodeling therapy in patients with stable hemodynamics.[17,168]

Recommendation 7.1

The panel suggests considering the risk-benefit ratio of LT in patients with hypertrophic cardiomyopathy with an elevated resting LVOT gradient (best practice statement, strong agreement).

Recommendation 7.2

The panel does not recommend considering an inducible LVOT obstruction seen with echocardiography as an absolute contraindication to LT in an asymptomatic patient (conditional recommendation, GRADE 2+, strong agreement).

Recommendation 7.3

The panel suggests not considering CMM as an absolute contraindication to LT (conditional recommendation, GRADE 2+, strong agreement).

Recommendation 7.4

The panel suggests carefully evaluating the risk-benefit ratio of LT in patients with severe systolic (especially those with ejection fraction <40%) and/or diastolic dysfunction (best practice statement, strong agreement).

Table 4
Expert-suggested research areas in cardiovascular assessment for liver transplant candidates in light of existing knowledge gaps.

Fields	Suggestions of future directions for research
General cardiovascular assessment of all LT candidates • Minimal cardiovascular examination • Preoperative iron and hemoglobin assessment and management • Preoperative outcome markers	• Artificial intelligence-based algorithms using ECG and TTE parameters determined during preoperative screening should be developed to predict perioperative complications. • High-quality studies are needed to assess the feasibility, efficacy, and safety of hemoglobin optimization before LT. • A comprehensive preoperative risk score incorporating clinical factors and biomarkers should be developed to predict postoperative cardiovascular morbidity and mortality.
LT candidate functional status assessment	• **A pragmatic and user-friendly tool for assessing frailty and physical capacity should be specifically developed and validated for LT candidates.** • High-quality studies are required to define threshold values in functional status assessments that could preclude candidacy for LT. • Specific prehabilitation pathways should be developed and prospectively validated.
Specific coronary artery disease assessment	• The role of coronary perfusion assessments that provide both functional and anatomical information, such as CT-derived fractional flow reserve (CT-FFR), requires validation in LT candidates. • A composite score based on parameters derived from clinical findings, cardiac biomarkers, and imaging techniques should be developed to improve cardiovascular screening in LT recipients. • Evidence is needed to determine the most appropriate timing and intervals for repeating clinical risk assessments and CTCA in LT candidates awaiting transplantation. • The cardiovascular evaluation approach should be optimized for use in resource-constrained settings.
Minimum cardiac evaluation in LT candidates hospitalized in the ICU with ACLF or ALF	• **There is a lack of studies evaluating cardiovascular risk profiles and management in patients with ACLF or ALF. Comprehensive analysis of pre- and postoperative cardiovascular risks and outcomes in these populations is needed.**
Specific interventions to perform before LT according to the patients coronary risk	• **Future randomized controlled trials should stratify LT candidates based on pre-existing cardiovascular risk factors and evaluate the impact of coronary revascularization on posttransplant outcomes to inform evidence-based guidelines.** • Emerging noninvasive cardiac imaging modalities should be investigated for their potential to shift the paradigm of preoperative evaluation toward personalized, risk-based revascularization strategies. • Biomarkers predicting the benefit of coronary revascularization in asymptomatic LT candidates need to be identified in order to develop guidelines to optimize pretransplant cardiac care.
Perioperative assessment or interventions to perform in LT candidates with atrial fibrillation	• As prospective RCTs on AF management in LT candidates are unlikely, future research should prioritize goal-directed medical therapy and the development of risk stratification tools to quantify the impact of AF on posttransplant outcomes.
Perioperative assessment or interventions to perform in LT candidates with valvular disease	• The long-term outcome of patients with percutaneous heart valve intervention compared with cardiac surgery in LT candidates should be prospectively assessed. • Prospective studies are needed to compare long-term outcomes of percutaneous vs surgical heart valve interventions in LT candidates to guide optimal treatment selection.

(*continued on next page*)

Table 4 (*continued*)

Fields	Suggestions of future directions for research
Perioperative assessment and/or interventions to perform in LT candidates with cardiomyopathy	• **As LT is increasingly performed in patients with cardiomyopathy, it is essential to evaluate short-term outcomes in this population using large, multicenter cohorts.** • Incorporating routine evaluation of cirrhotic cardiomyopathy could clarify its influence on perioperative outcomes and guide perioperative management strategies of LT candidates and recipients.

ACLF, acute-on-chronic liver failure; AF, atrial fibrillation; ALF, acute liver failure; CT, computed tomography; ECG, electrocardiogram; ICU, intensive care unit; LT, liver transplantation; RCT, randomized controlled trial; TTE, transthoracic echocardiogram.

3.16. Future directions for research on cardiovascular assessment in liver transplant Candidates

Our literature search highlighted a significant gap in current knowledge, which limited the ability of the panel to make more specific recommendations. To address this, the panel proposed several potential directions for future research, which are summarized in Table 4.

Declaration of competing interest

The authors of this manuscript have no conflicts of interest to disclose as described by *American Journal of Transplantation*.

Data availability

Not applicable.

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